

MATH 4720 / MSSC 5720

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Chapter 11



Department of Mathematical and Statistical Sciences

REGRESSION

- A regression function describes how a response variable y changes as an explanatory variable x changes.
- We often use a **regression line** to predict the value of y for a given value of x .
- Example: How much should you pay for a house?
- What factors are important in determining a reasonable price?
 - Amenities
 - Location
 - **Square footage**
- To determine a price, you might consider a model of the form:

$$\text{Price} = f(\text{square footage}) + \epsilon$$

EXAMPLE – AUSTRALIAN INSTITUTE OF SPORT



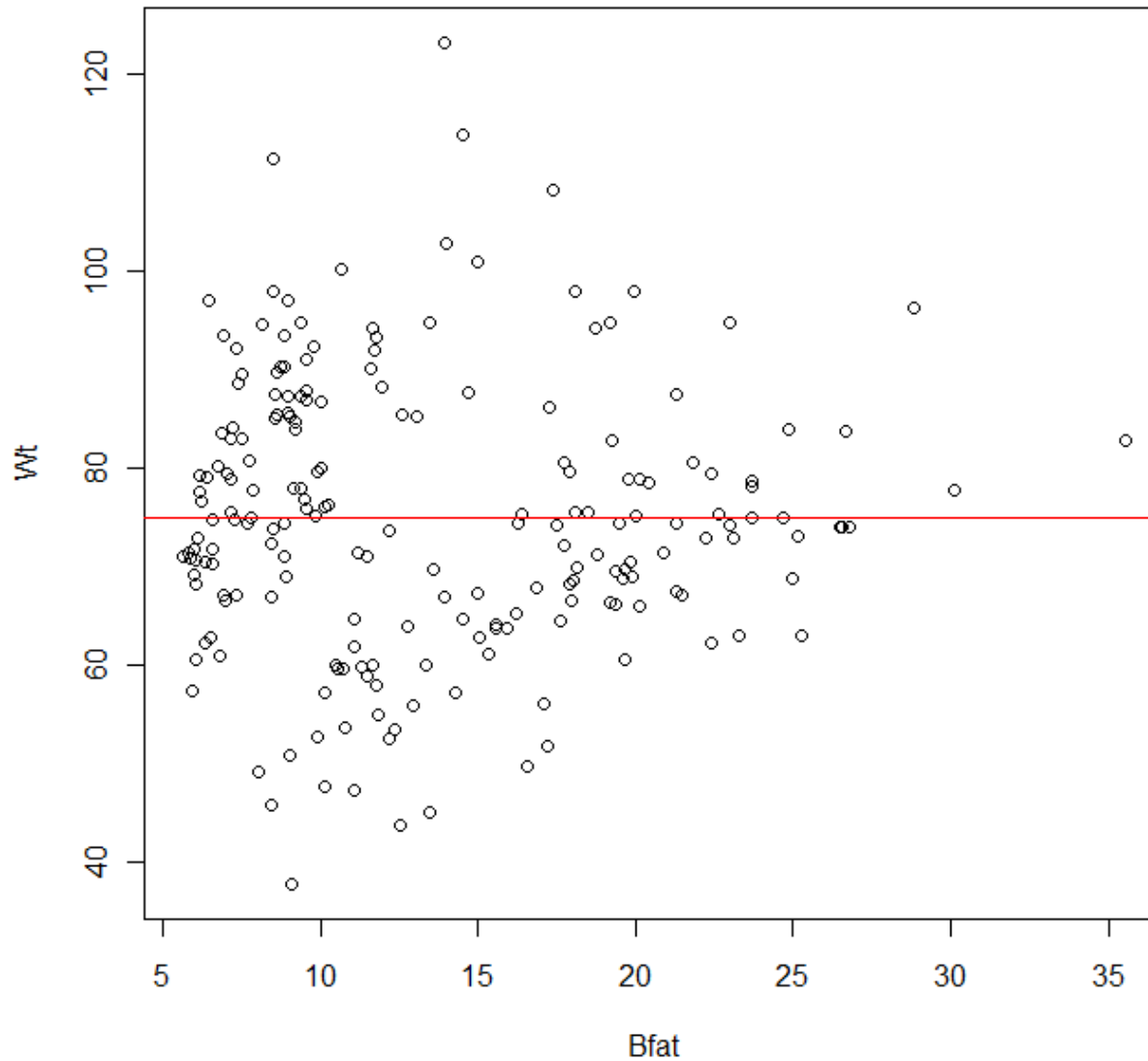
- **Data on 102 male and 100 female athletes collected at the Australian Institute of Sport, (courtesy of Richard Telford and Ross Cunningham.)**

	Gender	Bfat	Wt
• 1	female	19.75	78.9
• 2	female	21.30	74.4
• 3	female	19.88	69.1
• 4	female	23.66	74.9
• 5	female	17.64	64.6
	:	:	:
• 198	male	11.79	93.2
• 199	male	10.05	80.0
• 200	male	8.51	73.8
• 201	male	11.50	71.1
• 202	male	6.26	76.7

EXAMPLE CONT'D



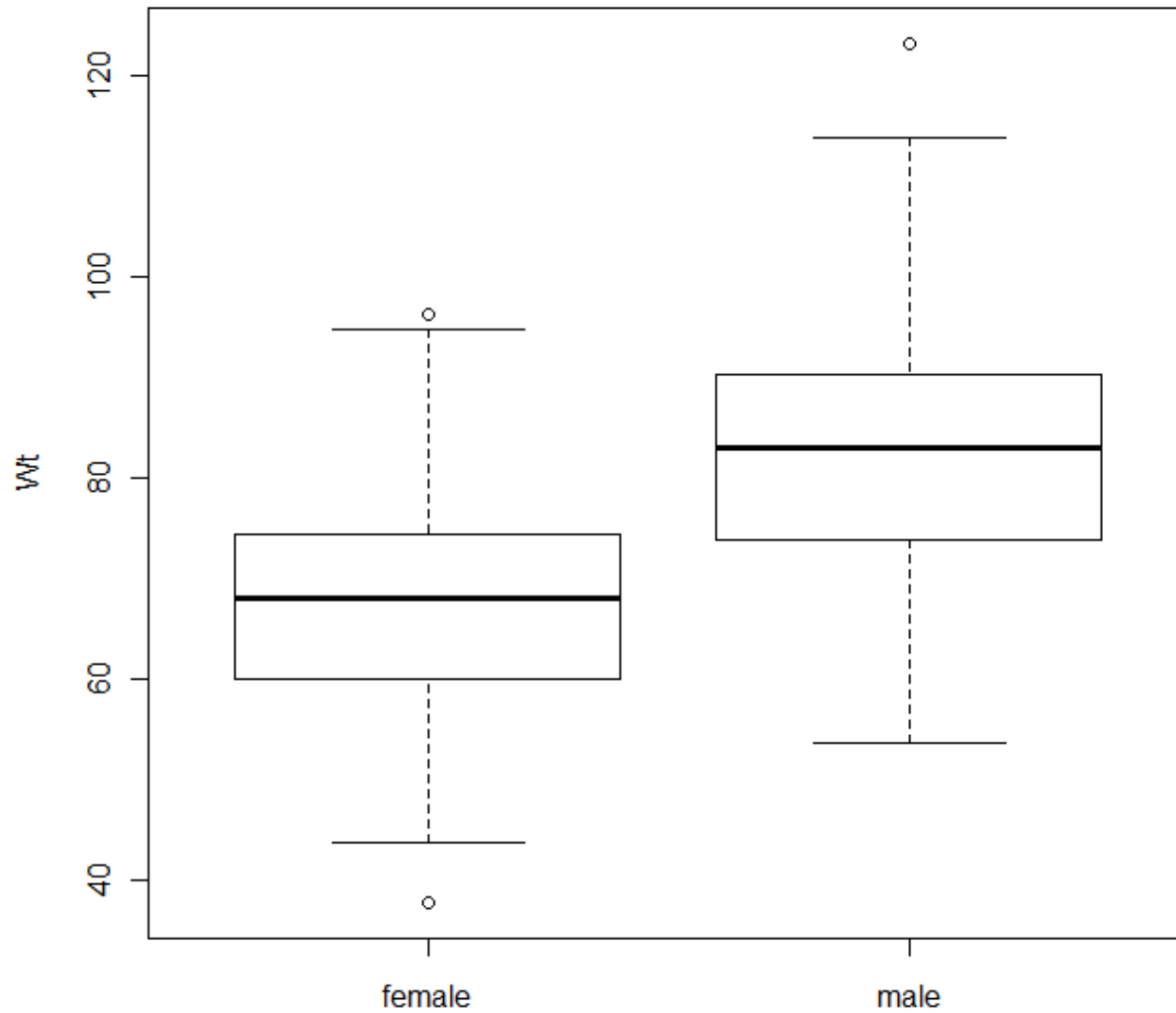
Australian Institute of Sport - Bfat vs Wt



EXAMPLE CONT'D



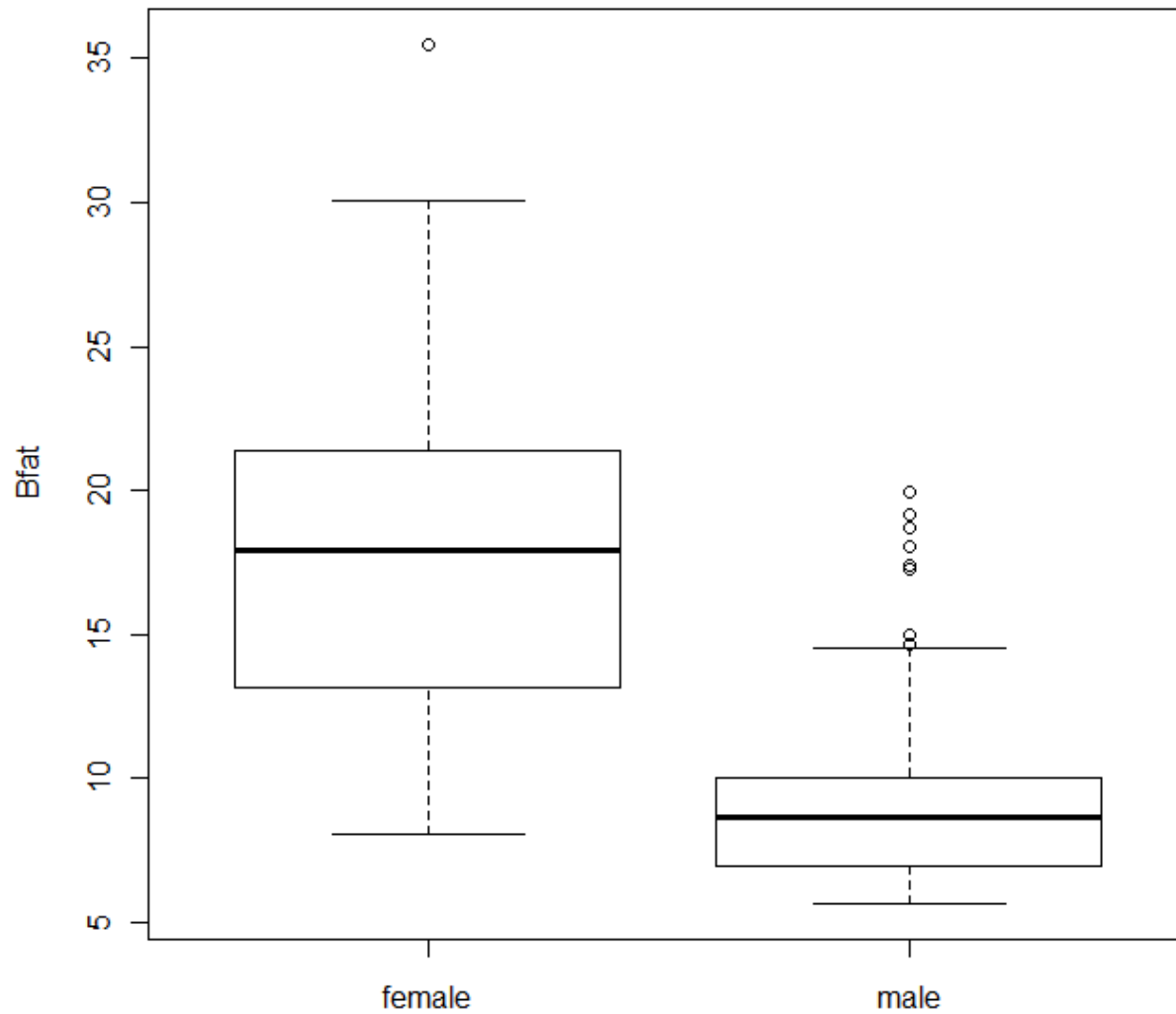
Australian Institute of Sport - Wt vs Gender



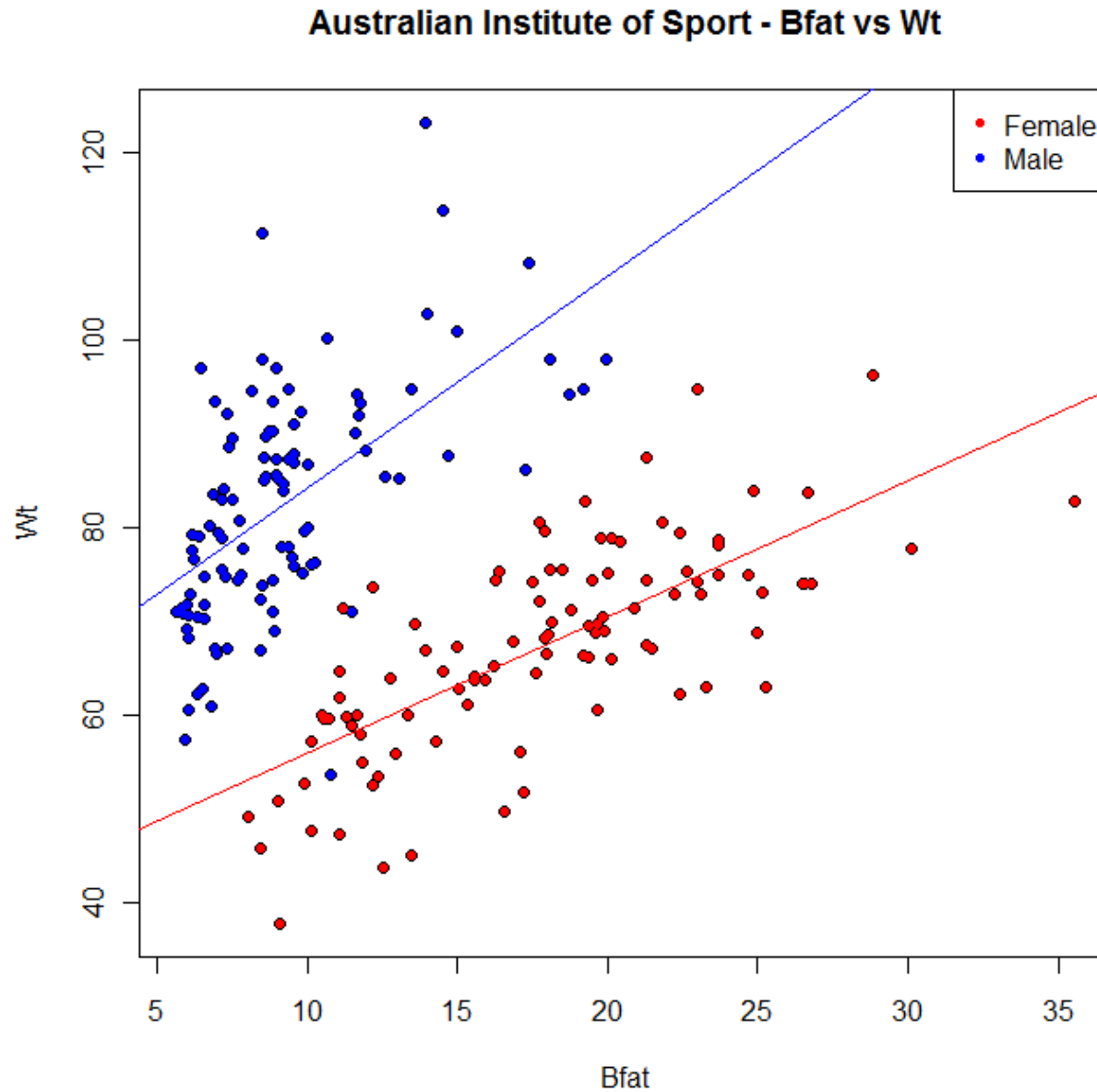
EXAMPLE CONT'D



Australian Institute of Sport - Bfat vs Gender



EXAMPLE CONT'D





REGRESSION LINE

- A **regression line** is a straight line that describes how a response variable y changes as an explanatory variable x changes.
- We often use a regression line to predict the value of y for a given value of x .
- Suppose that y is a response variable (plotted on the vertical axis) and x is an explanatory variable (plotted on the horizontal axis). A straight line relating y to x has an equation of the form

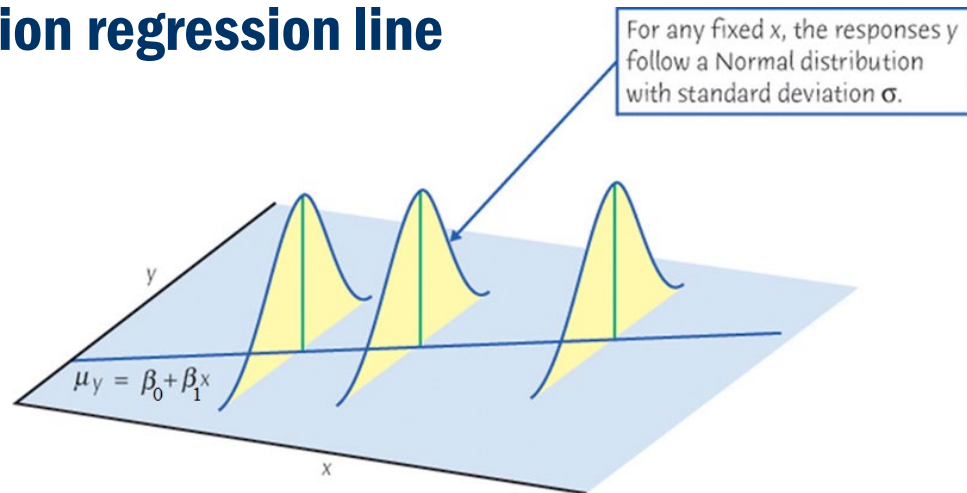
$$y = a + bx$$

- In this equation, b is the **slope**, the amount by which y changes when x increases by one unit. The number a is the **intercept**, the value of y when $x = 0$.

POPULATION REGRESSION LINE

- We often assume that for any fixed value of x , the response y varies according to a **Normal distribution**. Additionally, we assume that repeated responses y are independent of each other. The mean response μ_y has a straight-line relationship with x given by a population regression line

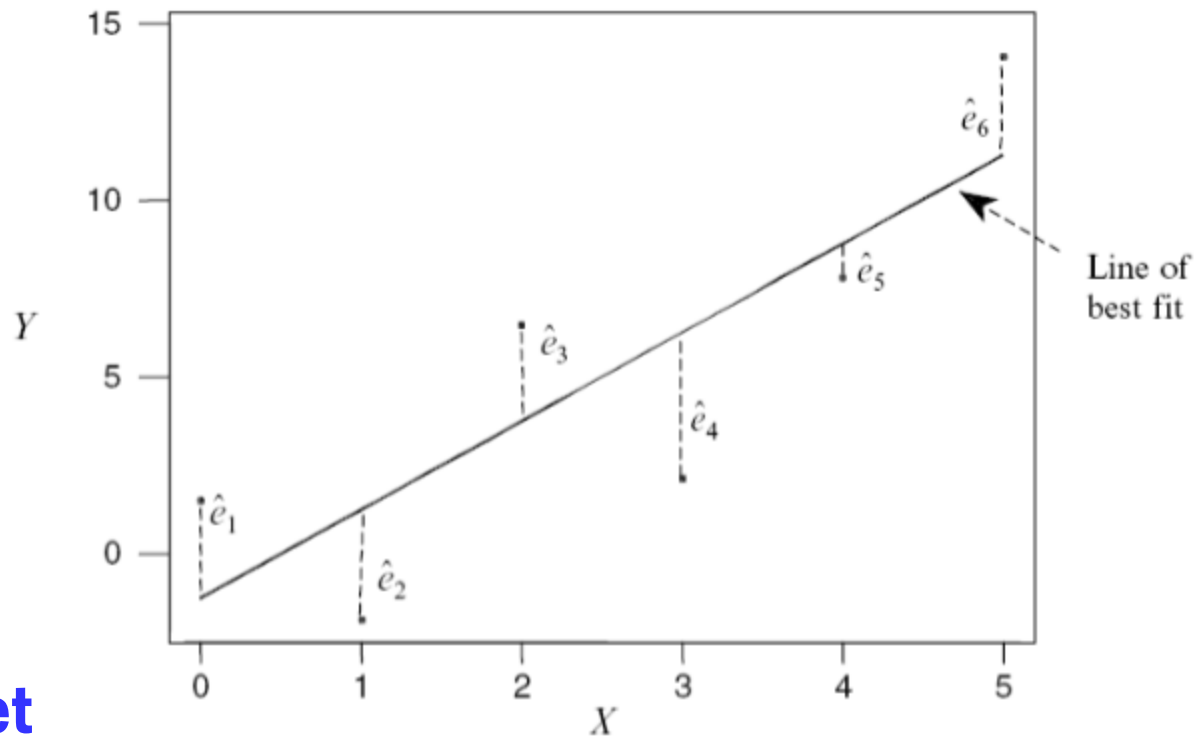
$$\mu_y = \beta_0 + \beta_1 x$$



- The **slope** β_1 and **intercept** β_0 are unknown parameters. The standard deviation of y at a given x (call it σ) is assumed to be the same for all values of x . The value of σ is unknown. There are thus three population parameters that we must estimate from the data: β_0 , β_1 and σ .

LEAST SQUARES REGRESSION LINE

- The least-squares regression line of y on x is the line that makes the sum of the squares of the vertical distances of the data points from the line as small as possible.
- **Residual:** $\hat{e}_i = y_i - \hat{y}_i$



- [Applet](#)

SIMPLE LINEAR REGRESSION

- The simplest model form to consider is

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

- Y_i is called the **dependent variable** or **response**.
- X_i is called the **independent variable** or **predictor**.
- ϵ_i is the random error term which is typically assumed to have
 - a Normal distribution with mean 0 and variance σ^2 .
 - We also assume that error terms are independent of each other.

LEAST SQUARES CRITERION

- If the simple linear model is appropriate then we need to estimate the values β_0 and β_1 .
- To determine the line that best fits our data, we choose the line that minimizes the sum of squared vertical deviations from our observed points to the line.
- In other words, we minimize

$$Q = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

LEAST SQUARES ESTIMATORS

- **Objective Function:**

- $Q = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$

- **Take the derivatives with respect to β_0 :**

- $\frac{dQ}{d\beta_0} = \sum_{i=1}^n \frac{d}{d\beta_0} (Y_i - \beta_0 - \beta_1 X_i)^2$
 $= -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)$

- **Setting the derivative equal to zero:**

- $\sum_{i=1}^n Y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^n X_i = 0$
- $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

- **To estimate β_1 , likewise:**

- $\frac{dQ}{d\beta_1} = \sum_{i=1}^n \frac{d}{d\beta_1} (Y_i - \beta_0 - \beta_1 X_i)^2$
 $= -2 \sum_{i=1}^n X_i (Y_i - \beta_0 - \beta_1 X_i)$

LEAST SQUARES ESTIMATORS

- **Setting the derivative $\frac{dQ}{d\beta_1}$ equal to zero and plugging in $\hat{\beta}_0$:**
 - $\sum_{i=1}^n X_i Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X})n\bar{X} - \hat{\beta}_1 \sum_{i=1}^n X_i^2 = 0$
 - $\hat{\beta}_1 = \frac{\sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^n X_i^2 - n\bar{X}^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$
- **Letting**
 - $S_{xy} = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$
 - $S_{xx} = \sum_{i=1}^n (X_i - \bar{X})^2$
 - **We have:** $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$
- **How about error variance σ^2 ?**
 - **Use the sum of squared deviations of the points from the regression line:**
 - $\hat{\sigma}^2 = MSE = \frac{SSE}{df_E} = \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n-2}$
 - **where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ and**
 - **n is the # of pairs (X_i, Y_i)**

LEAST SQUARES ESTIMATORS

- For n pairs of (X_i, Y_i) s, where $i = 1, \dots, n$
- If $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$, where ϵ_i s are independent $N(0, \sigma^2)$
- Let's define $\bar{X} = 1/n \sum_i X_i$ and $\bar{Y} = 1/n \sum_i Y_i$
 - $S_{xx} = \sum_{i=1}^n (X_i - \bar{X})^2$
 - $S_{xy} = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$
- We can estimate the regression coefficients as:
 - $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$
 - $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$
- Furthermore, let's define the predicted value, $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$
 - $SS_E = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$ and $S_{yy} = \sum_{i=1}^n (Y_i - \bar{Y})^2$
- We have the following:
 - $\hat{\sigma}^2 = \frac{SS_E}{n-2}$
 - $r_{XY} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$, r_{XY} is correlation coefficient (were discussed in chapter 3)

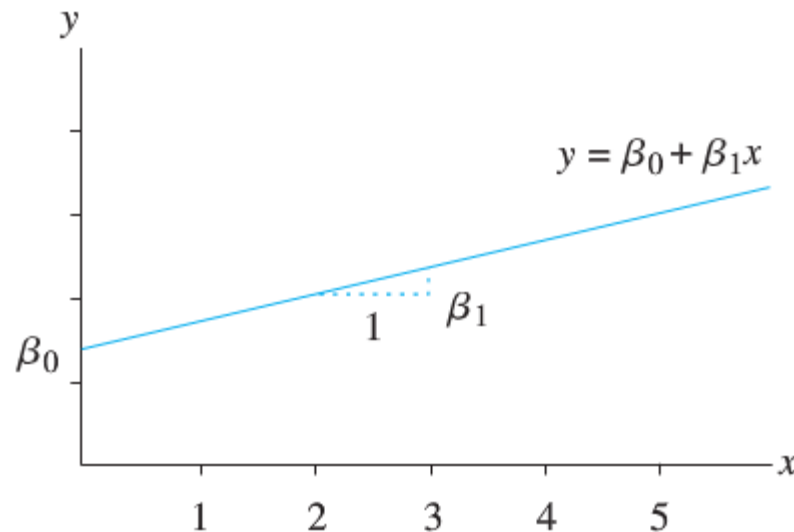


RELATIONSHIPS BETWEEN 2 NUMERIC VARIABLES

- **Correlation** or **r** : measures the direction and strength of the linear relationship between two numeric variables
 - General Properties
 - It must be between -1 and 1, or $(-1 \leq r \leq 1)$.
 - If r is negative, the relationship is negative.
 - If $r = -1$, there is a perfect negative linear relationship (extreme case).
 - If r is positive, the relationship is positive.
 - If $r = 1$, there is a perfect positive linear relationship (extreme case).
 - If r is 0, there is no **linear** relationship.
 - r measures the strength of the **linear** relationship.
 - If explanatory and response are switched, r remains the same.
 - r has no units of measurement associated with it
 - Scale changes do not affect r
- **Correlation Applet**

INFERENCE

- Interpretation of the parameters for $Y = \beta_0 + \beta_1 X_i$
- Often times, inference for the slope parameter, β_1 , is most important.
- β_1 tells us the expected change in Y per unit change in X .



- Note that
 - β_0 is the expected value of y when $x = 0$
 - β_1 is the expected rate of change,
 - $\beta_0 = E(y|x = 0)$, and $\beta_1 = \frac{d}{dx} E(y|x)$

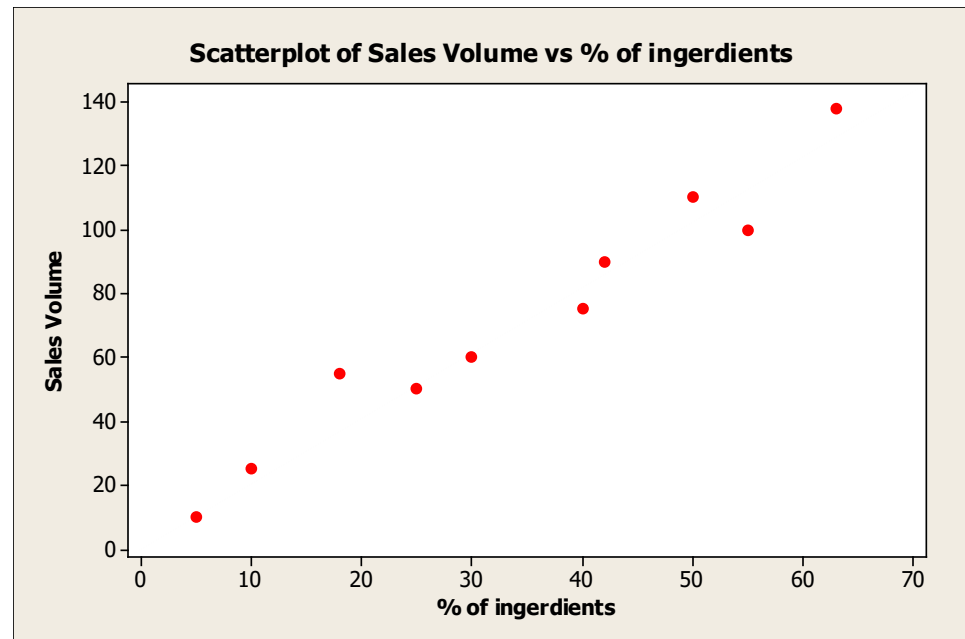
INFERENCE CONT'D

- In practice, it may be of interest to estimate β_0 and β_1 . Estimation of β_1 is more important.
 - If we conclude that β_1 equals 0, then we are concluding that there is no linear relationship between Y and X .
 - If we conclude that β_1 equals 0, then it makes no sense to use our linear model with X to predict Y .
- The Confidence intervals of β_0 and β_1 are given by
 - $\hat{\beta}_1 \pm t_{\alpha/2} se(\hat{\beta}_1)$ (df = n-2)
 - where $se(\hat{\beta}_1) = \sqrt{\frac{MSE}{S_{xx}}}$
 - $\hat{\beta}_0 \pm t_{\alpha/2} se(\hat{\beta}_0)$ (df = n-2)
 - where $se(\hat{\beta}_0) = \sqrt{MSE \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}} \right)}$

BOOK EXAMPLE 11.2:

- **Data from a sample of 10 pharmacies are used to examine the relation between prescription sales volume and the percentage of prescription ingredients purchased directly from the supplier.**

Pharmacy	Sales Volume, y (in \$1,000)	% of Ingredients Purchased Directly, x
1	25	10
2	55	18
3	50	25
4	75	40
5	110	50
6	138	63
7	90	42
8	60	30
9	10	5
10	100	55



BOOK EXAMPLE 11.2 CONT'D



- $\bar{X} = 1/n \sum_i X_i = 33.8$
- $\bar{Y} = 1/n \sum_i Y_i = 71.3$
- $S_{xx} = \sum_{i=1}^n (X_i - \bar{X})^2 = 3407.6$
- $S_{xy} = \sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = 6714.6$

	y	x	y - $\bar{y}$	x - $\bar{x}$	(x - $\bar{x}$)(y - $\bar{y}$)	(x - $\bar{x}$) ²
	25	10	-46.3	-23.8	1,101.94	566.44
	55	18	-16.3	-15.8	257.54	249.64
	50	25	-21.3	-8.8	187.44	77.44
	75	40	3.7	6.2	22.94	38.44
	110	50	38.7	16.2	626.94	262.44
	138	63	66.7	29.2	1,947.64	852.64
	90	42	18.7	8.2	153.34	67.24
	60	30	-11.3	-3.8	42.94	14.44
	10	5	-61.3	-28.8	1,765.44	829.44
	100	55	28.7	21.2	608.44	449.44
Totals	713	338	0	0	6,714.60	3,407.60
Means	71.3	33.8				

- $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{6714.6}{3407.6} = 1.97$
- $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 71.3 - 1.97(33.8) = 4.70$
- **Prediction formula for X:** $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X = 4.70 + 1.97(X)$

➤ **In R:** `summary(lm(y ~ x, data=exmp11.2))`

Confidence Intervals:

- $\hat{\beta}_0 \pm t_{\alpha/2} se(\hat{\beta}_0)$ (df = 8)
 - $4.70 \pm 2.306 (5.95)$
- $\hat{\beta}_1 \pm t_{\alpha/2} se(\hat{\beta}_1)$ (df = 8)
 - $1.97 \pm 2.306 (0.15)$

Regression Analysis: Sales Volume versus % of ingerdients

The regression equation is
Sales Volume = 4.70 + 1.97 % of ingerdients

Predictor	Coef	SE Coef	T	P
Constant	4.698	5.952	0.79	0.453
% of ingerdients	1.9705	0.1545	12.75	0.000

T-statistics and p-value for testing
 $H_0: \beta_1 = 0$ vs $H_a: \beta_1 \neq 0$

HYPOTHESIS TESTING

- $H_0: \beta_1 = 0$ (This means that y does not depend on x)
 - $H_a: \beta_1 \neq 0$ (This means that y depends on x)
 - **T.S.** $t = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)}$
 - **Reject H_0 in favor of H_a if $|t| > t_{\alpha/2}$ (df = n-2)**
-

- $H_0: \beta_0 = 0$ (This means that $E(y|x = 0) = 0$)
- $H_a: \beta_0 \neq 0$ (This means that $E(y|x = 0) \neq 0$)
- **T.S.** $t = \frac{\hat{\beta}_0}{se(\hat{\beta}_0)}$
- **Reject H_0 in favor of H_a if $|t| > t_{\alpha/2}$ (df = n-2)**

BACK TO EXAMPLE 11.2

- Suppose, we want to test if the Sale volume depend on the % of ingredients:

- Minitab Output

Regression Analysis: Sales Volume versus % of ingerdients

The regression equation is

Sales Volume = 4.70 + 1.97 % of ingerdients

Predictor	Coef	SE Coef	T	P
Constant	4.698	5.952	0.79	0.453
% of ingerdients	1.9705	0.1545	12.75	0.000

- $H_0: \beta_1 = 0$ vs. $H_a: \beta_1 \neq 0$
- **T.S.** $T = 12.75$ with $p\text{-value} = 0.000$
- **Conclusion:** Is $p\text{-value} < 0.05$? Yes. We reject H_0 in favor of H_a . We have sufficient evidence to conclude that the Sale volume depend on the % of ingredients.

ASSUMPTIONS

- For both confidence interval and hypothesis testing problems, we make assumptions on the model

- $$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n$$

- **Assumption 1:** y and x are linearly related. If not, some transformation is needed.

- y and x are linearly related can be checked through scatter plot.

- **Book**
- **Example 11.1:**
- **Free Flights**

FIGURE 11.7

Frequent flyer free flights
by month

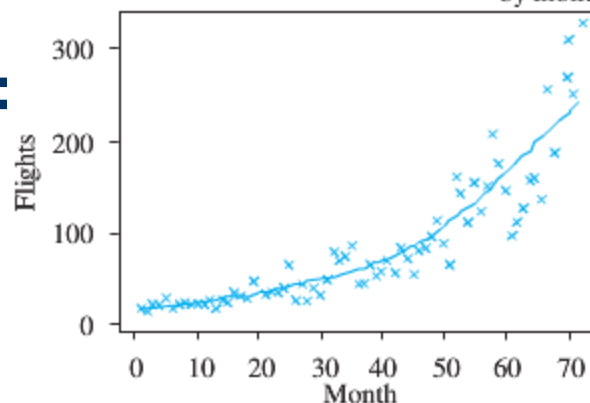
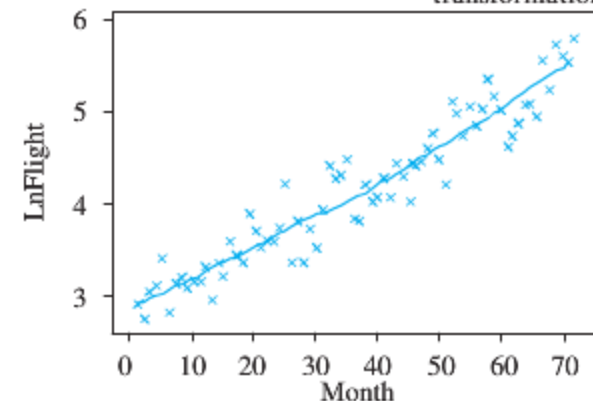


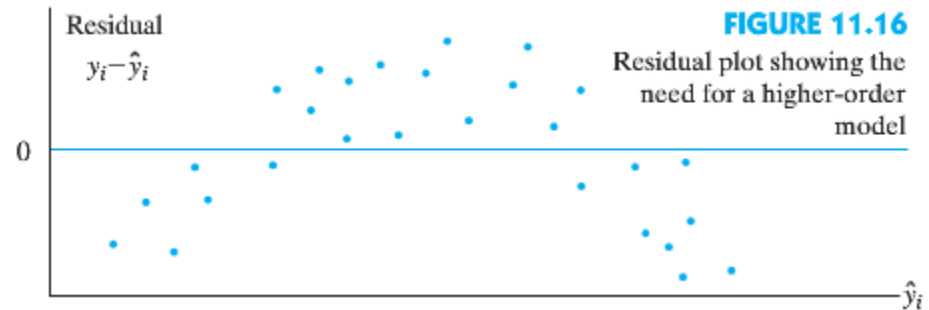
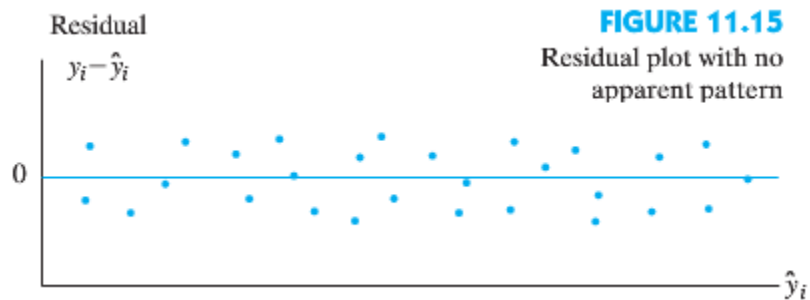
FIGURE 11.8

Result of logarithm
transformation

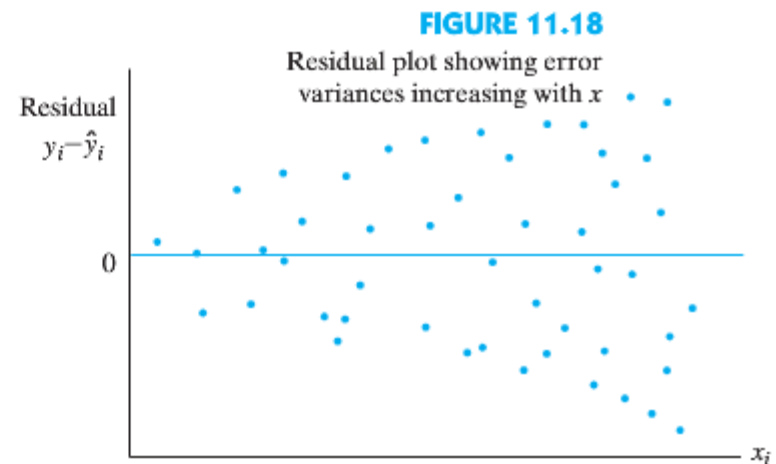
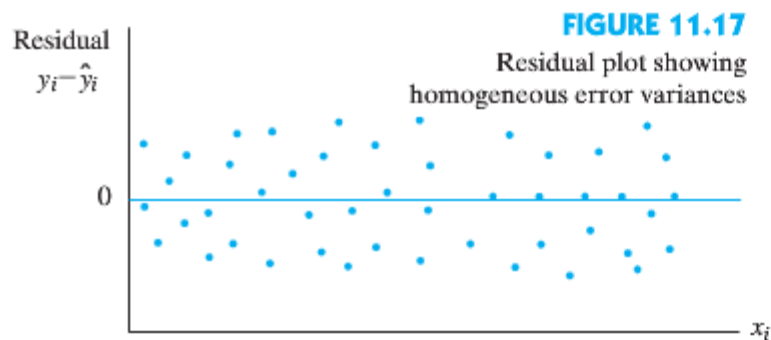


HOW TO CHECK THE ASSUMPTIONS

- **Assumption 2: The error terms $\epsilon_i, i = 1, 2, \dots, n$ are independent and identically distributed as normal.**
 - Second assumption that errors are independent and identically distributed can be checked through the normal probability plot of the residual, and the residual plots.



- **Assumption 3: $Var(\epsilon_i) = constant$**



4 DATASETS WITH THE SAME LEAST SQUARES REGRESSION LINE

- 4 Scatterplot with same Regression Estimates:

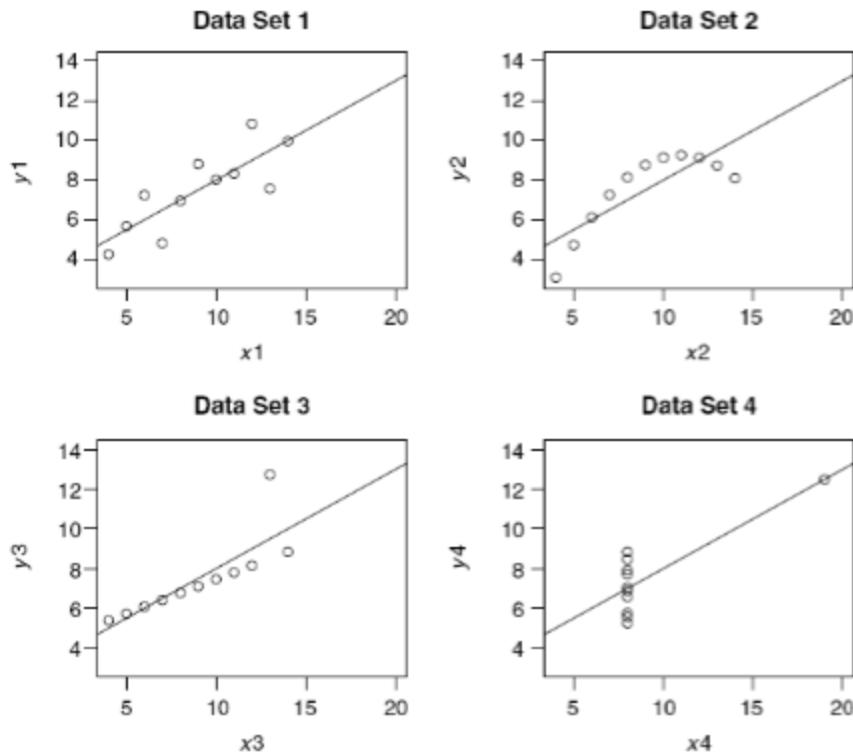
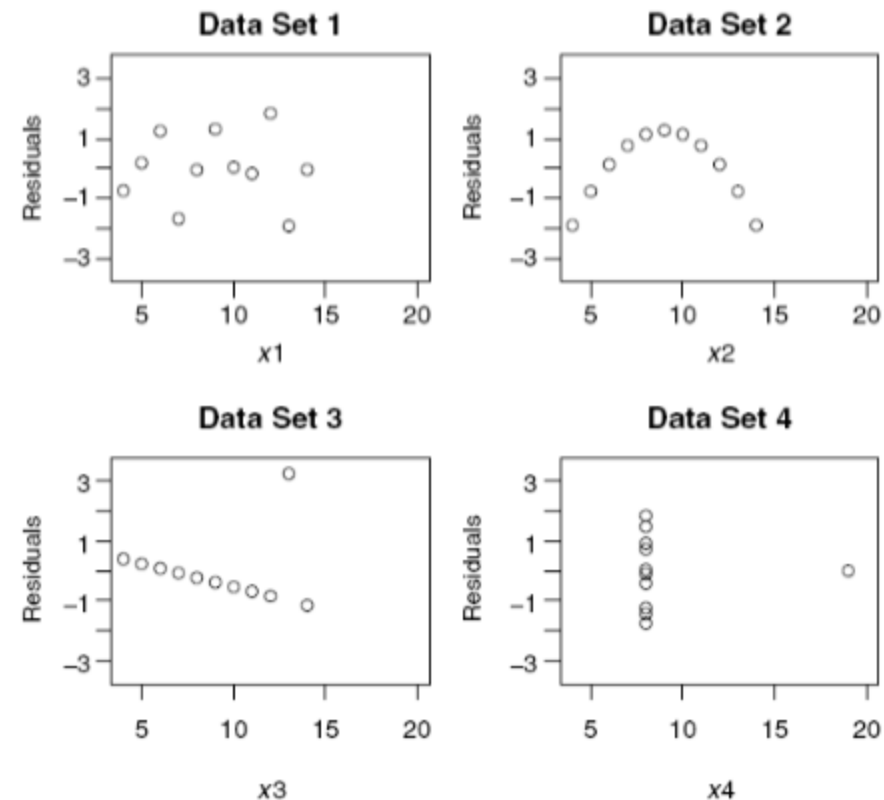


Figure 3.1 Plots of Anscombe's four data sets

- 4 Residual Plots:



- Source: Sheather, S.J. (2009) A Modern Approach to Regression with R, Springer, New York

REGRESSION ANALYSIS USING ANOVA



- **Note that there is variation in y :**

- some of it is due to regression: SS_{Reg}
- and some due to error: SS_E

- $$SS_{Total} = SS_{Reg} + SS_E$$
$$df_{Total} = df_{Reg} + df_E$$

- **If Regression is significant, then**

$$F = \frac{SS_{Reg}/df_{Reg}}{SS_E/df_E} > F_{\alpha}(df_1 = 1, df_2 = n - 2)$$

- **R Output:**

Source	DF	SS	MS	F-ratio
Regression/ Explained	1	SS_{Reg}	$MS_{Reg} = SS_{Reg}/1$	$F = MS_{Reg}/MSE$
Residual/Error /Unexplained	n-2	SS_E	$MSE = SS_E/(n-2)$	
Total	n-1 = $df_{Reg} + df_E$	$SST =$ $SS_{Reg} + SS_E$		

COEFFICIENT OF DETERMINATION

- One way to answer the question of reliability is through a **coefficient of determination (R^2)**
- R^2 = Proportion of Variability in Y due to Regression
$$= \frac{SS_{Reg}}{SS_{Tot}} = 1 - \frac{SS_E}{SS_{Tot}}$$
- **Reminder:** $SS_{Total} = SS_{Reg} + SS_E$
- R is also the correlation in simple linear regression.
- If $R^2 \approx 1$, then most of the variability can be attributed to Regression. In this case, prediction is reliable.
- If $R^2 \approx 0$, then most of the variability is due to error. In this case prediction is not reliable.

PREDICTION

- **Probably the most important objective of regression is to predict Y for a given X .**
- **Based on the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, Y can be predicted as**

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

- **Back to Example 11.2: Suppose we want to predict sale volume for a pharmacy that purchases 15% of its prescription ingredients directly from the supplier.**
- **Prediction formula for X:**
 - $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X = 4.70 + 1.97(X)$
 - $\hat{Y} = 4.70 + 1.97(15) = 34.26$

CONFIDENCE INTERVAL AND PREDICTION INTERVAL

- **100% Confidence Interval of $E(y|x^*) = \beta_0 + \beta_1 x^*$**
 - $E(\widehat{y|x^*}) \pm t_{\alpha/2} * se(E(\widehat{y|x^*}))$, **where**
 - $E(\widehat{y|x^*}) = \hat{\beta}_0 + \hat{\beta}_1 x^*$
 - $se(E(\widehat{y|x^*})) = \sqrt{MSE \left\{ \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}} \right\}}$
- **100% Prediction Interval of $y = \beta_0 + \beta_1 x^* + \epsilon$**
 - $\hat{y} \pm t_{\alpha/2} * se(\hat{y})$, **where**
 - $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x^*$
 - $se(\hat{y}) = \sqrt{MSE \left\{ 1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}} \right\}}$
- **Main Difference:**
 - the confidence interval $E(y|x)$ provides the interval of the **average of y** , while
 - the prediction interval of y provides the interval of the **individual y** .

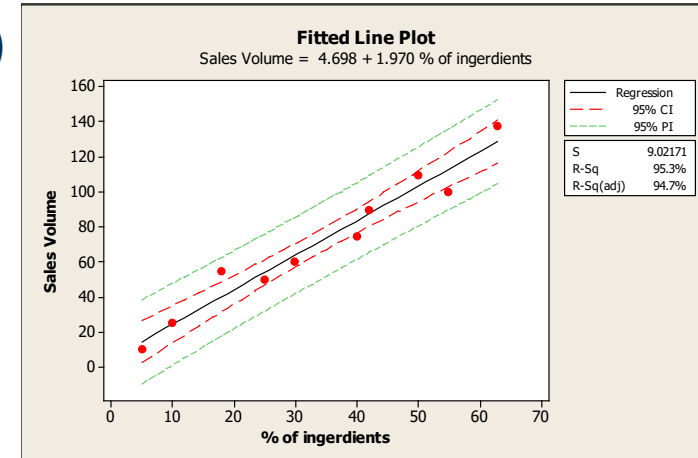
BACK TO BOOK EXAMPLE 11.2:

- Suppose we want to obtain the **Confidence and Prediction intervals** of sale volume for a pharmacy that purchases 15% of its prescription ingredients directly from the supplier.
- In R:

```
➤ new <- data.frame(x = 15)
➤ predict(model, new, interval=c("confidence"))
      fit      lwr      upr
1 34.25502 24.86499 43.64505
➤ predict(model, new, interval=c("prediction"))
      fit      lwr      upr
1 34.25502 11.42996 57.08008
```

- Prediction band in R using ggplot:

```
➤ pred.int <- predict(model, interval = "prediction")
➤ mydata <- cbind(exmp11.2, pred.int)
# 2. Regression line + confidence intervals
➤ library("ggplot2")
➤ p <- ggplot(mydata, aes(x, y)) + geom_point() + stat_smooth(method = lm)
# 3. Add prediction intervals
➤ p + geom_line(aes(y = lwr), color = "red", linetype = "dashed")+
  geom_line(aes(y = upr), color = "red", linetype = "dashed")
```



INFLUENTIAL POINTS – BAD LEVERAGE POINTS

- An observation is **influential** for a statistical calculation if removing it would markedly change the result of the calculation.
- Points that are outliers in the x direction are often influential for the least-squares regression line.

- [Applet](#)

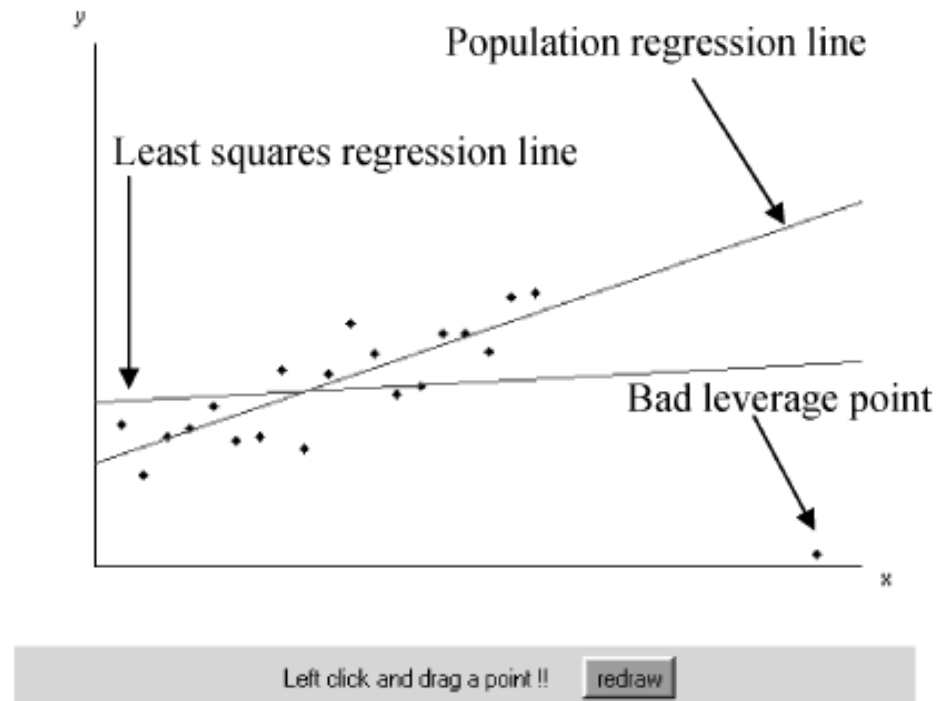
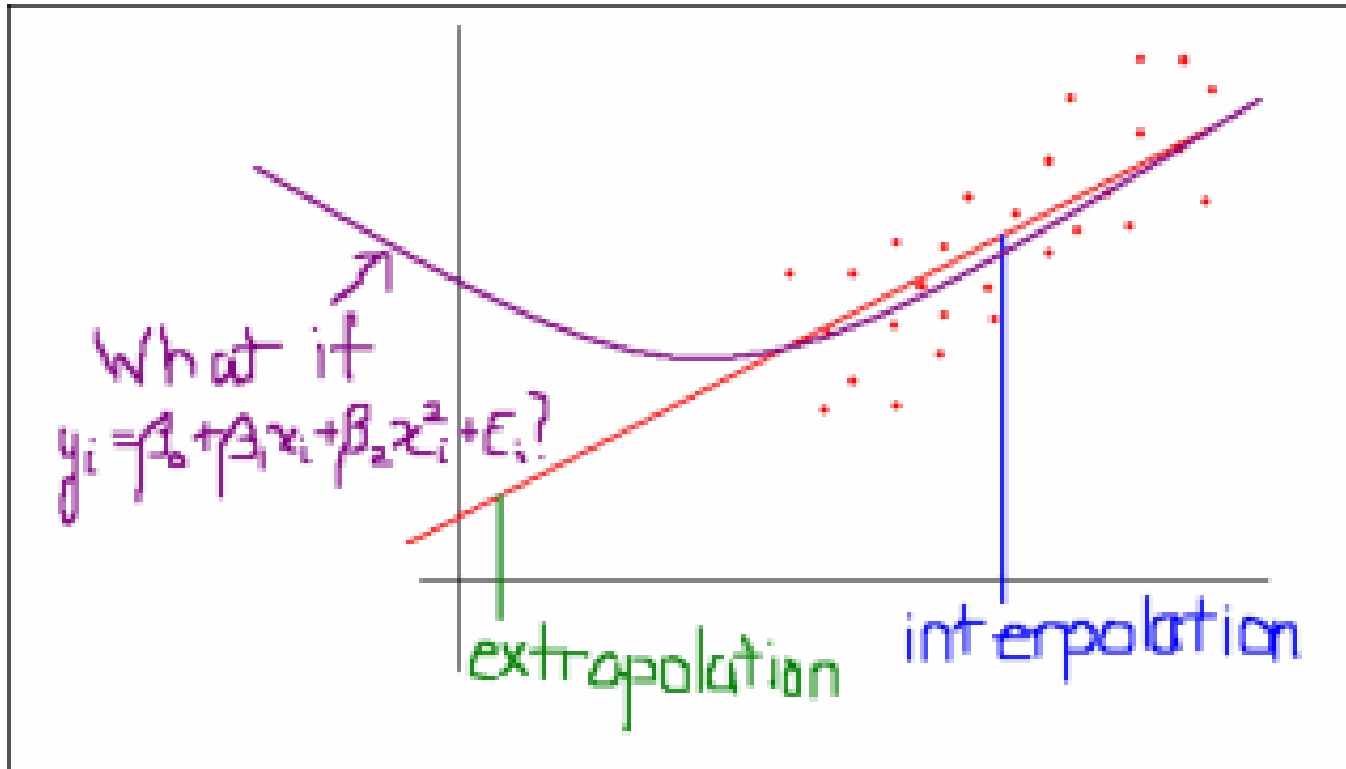


Figure 3.5 A plot showing a bad leverage point

EXTRAPOLATION VS INTERPOLATION



GOING BACK TO AIS DATA

- **Wrong Analysis:**

- **Fail to Reject**

- $H_0: \beta_1 = 0$

- $R^2 \approx 0,$

- **Most of the variability is due to Error**

Regression Analysis: Wt versus Bfat

The regression equation is
Wt = 75.0 - 0.000 Bfat

Predictor	Coef	SE Coef	T	P
Constant	75.013	2.363	31.75	0.000
Bfat	-0.0004	0.1591	-0.00	0.998

S = 13.9603 R-Sq = 0.0% R-Sq(adj) = 0.0%

- **Regression Model is NOT significant**

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	0.0	0.0	0.00	0.998
Residual Error	200	38978.2	194.9		
Total	201	38978.2			

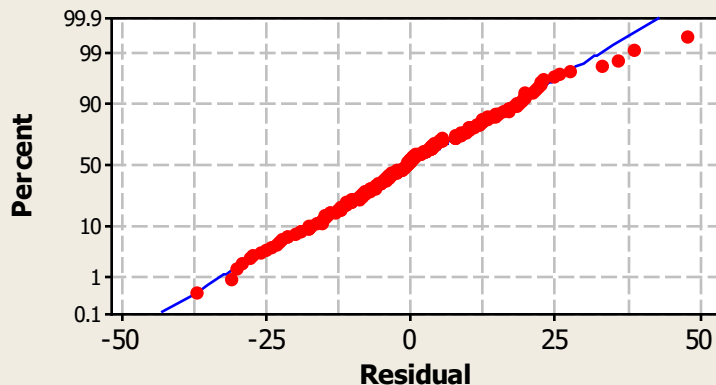
MODEL ASSUMPTIONS (WRONG ANALYSIS)



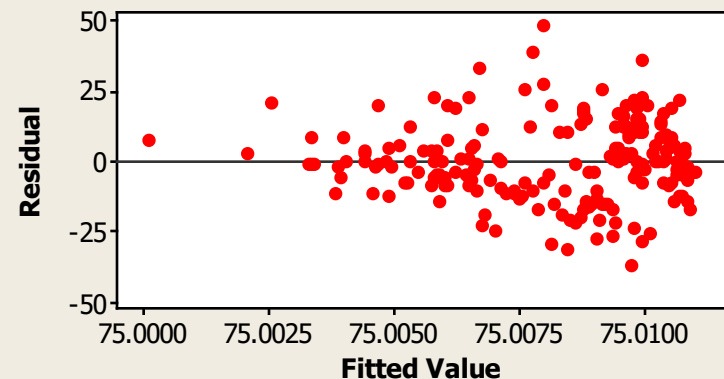
- The error terms $\epsilon_i, i = 1, 2, \dots, n$ are NOT independent
- $Var(\epsilon_i)$ is NOT constant

Residual Plots for Wt

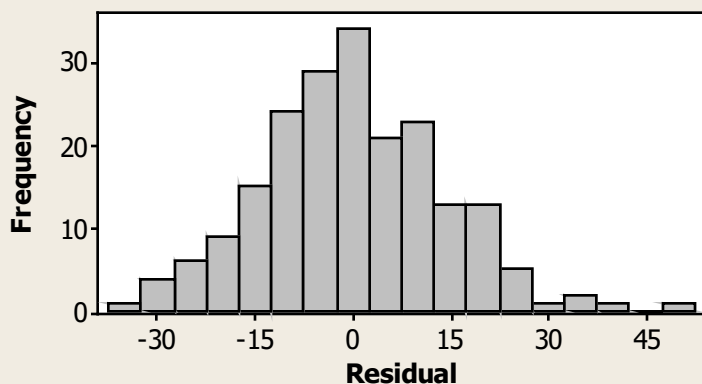
Normal Probability Plot



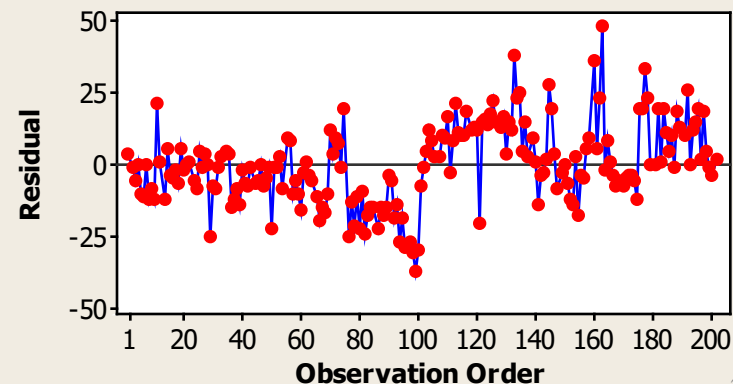
Versus Fits



Histogram



Versus Order



GOING BACK TO AIS DATA

- **Correct Analysis for “Gender=Female”**

- **Reject**

– $H_0: \beta_1 = 0$

Regression Analysis: Wt_F versus Bfat_F

The regression equation is
Wt_F = 41.4 + 1.45 Bfat_F

- $R^2 \approx 53\%$,

- **53% of the variability is due to Model**

Predictor	Coef	SE Coef	T	P
Constant	41.443	2.599	15.95	0.000
Bfat_F	1.4510	0.1393	10.42	0.000

S = 7.55769 R-Sq = 52.5%

R-Sq(adj) = 52.1%

- **Regression Model is significant**

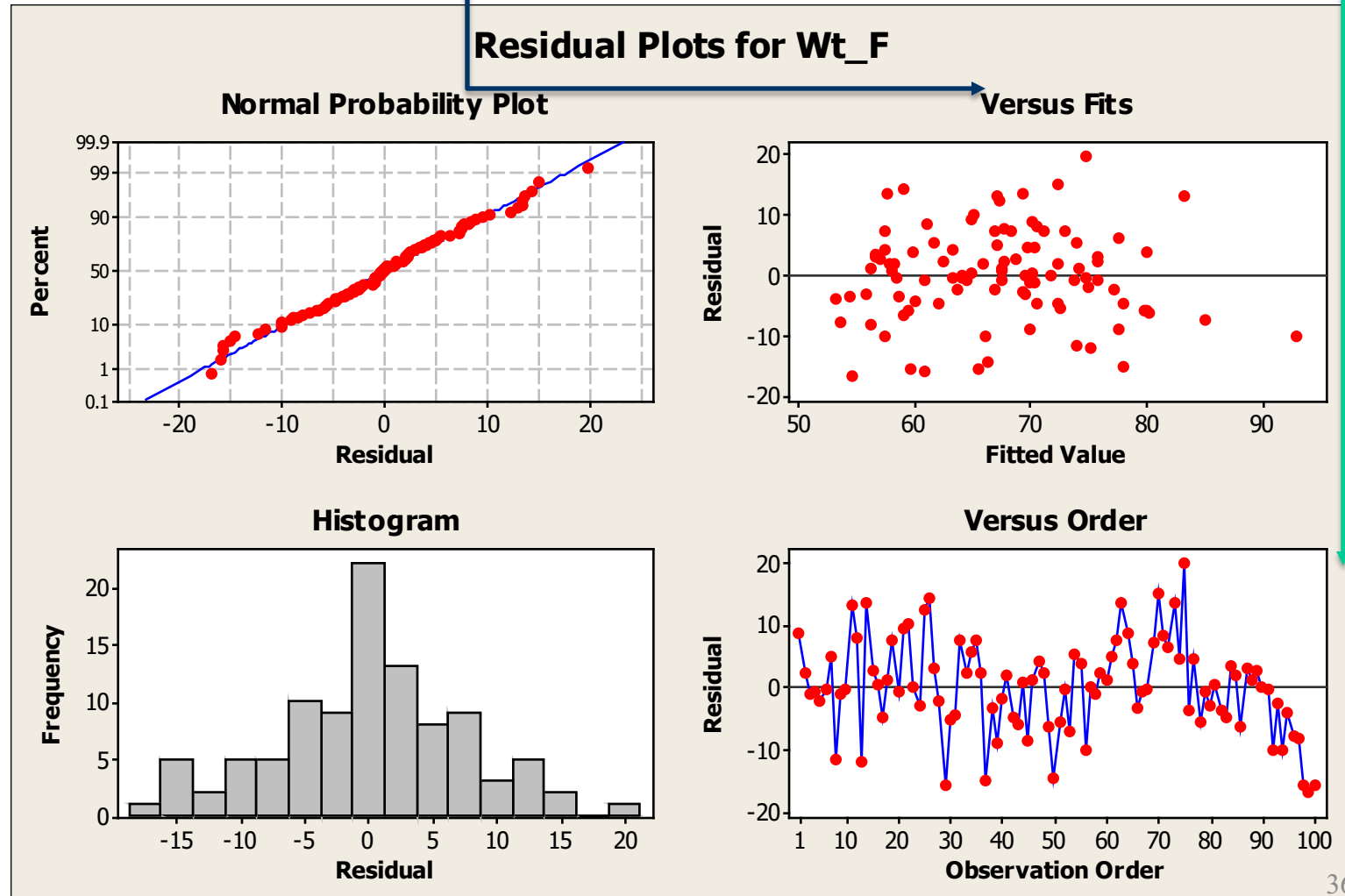
Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	6197.9	6197.9	108.51	0.000
Residual Error	98	5597.6	57.1		
Total	99	11795.6			

- $F = T^2$

MODEL ASSUMPTIONS

- The error terms $\epsilon_i, i = 1, 2, \dots, n$ are independent
- $Var(\epsilon_i)$ is constant



CAUTIONS ABOUT REGRESSION

- Regression is a powerful tool for describing the relationship between two variables. When you use these tools, you must be aware of their limitations.
- **Regression lines describe only linear relationships.** You can do the calculations for any relationship between two quantitative variables, but the results are useful only if the scatterplot shows a linear pattern.
- **Least-squares regression lines are not resistant.** Always plot your data and look for observations that may be influential.
- **Beware extrapolation.** Extrapolation is the use of a regression line for prediction far outside the range of values of the explanatory variable x that you used to obtain the line. Such predictions are often not accurate.